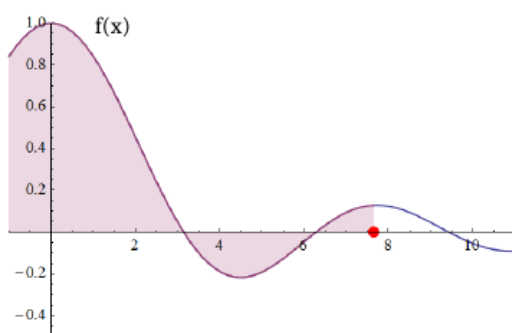


Problem 6

Problem 6

Problem 1 Given the following graph of $y = f(x)$, let $g(x) = \int_{-1}^x f(t) dt$.



- (a) Is g continuous? Why or why not?
- (b) Is g differentiable? Why or why not?
- (c) Where does g achieve its absolute maximum and minimum values? Where does g achieve any local extreme values? Assume the domain of g is $[-1, 7.6]$.
- (d) Where is the graph of g concave up? Concave down?